

ECON 101: Macroeconomics

Quiz 4

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Multiple Choice (1 mark each)

1. **Which of the following best describes the rule of 70?**
 - A. The number of years required for real GDP to double ≈ 70 divided by the growth rate of real GDP per capita.
 - B. Real GDP grows by 70 percent every year.
 - C. The price level doubles every 70 years.
 - D. Productivity rises by 70 percent when labour input doubles.

2. **Which of the following would *most directly* raise labour productivity in the long run?**
 - A. A temporary tax rebate to consumers.
 - B. Increased spending on worker training and education.
 - C. A higher nominal wage.
 - D. An increase in the money supply.

3. **If the Canadian government implements policies that increase national saving, in the long run we would expect:**
 - A. Lower investment and slower growth.
 - B. Higher investment and faster growth.
 - C. No change in potential output.
 - D. Higher consumption and lower investment.

4. **If planned aggregate expenditure is less than actual real GDP, firms will ...**
 - A. Increase production.
 - B. Decrease inventories.
 - C. Decrease production.
 - D. Raise prices to clear excess demand.

5. Which of the following would *decrease* planned investment spending?

- A. A decline in the real interest rate.
- B. A rise in expected future profitability.
- C. A deterioration in business confidence.
- D. An improvement in technology.

Problem 1 (6 marks)

Suppose a country's real GDP per capita is \$50,000 and grows at 2 percent per year.

- (a) Using the rule of 70, approximate how many years it will take for real GDP per capita to double.
- (b) If productivity growth accelerates to 3 percent, how many years will doubling now take?
- (c) Explain in one or two sentences why even a small change in the growth rate matters for living standards over generations.

Problem 2 (9 marks)

The following table shows a simplified economy (all figures in billions):

Component	Equation or Value
Consumption	$C = 250 + 0.8Y_d$, $Y_d = Y - T$
Planned Investment	$I = 150$
Government Purchases	$G = 200$
Net Exports	$NX = -50$
Taxes	$T = 100$

- (a) Compute planned aggregate expenditure (AE) as a function of Y .
- (b) Find the equilibrium level of output where $AE = Y$.
- (c) At this equilibrium, determine household saving.
- (d) Briefly interpret whether the government is running a budget surplus or deficit at this output level.